

Comprehensive PyTorch Cheatsheet

1. Tensor Creation

Operation	Code Example
Create a tensor from a list	<code>torch.tensor([1, 2, 3])</code>
Create a tensor of zeros	<code>torch.zeros(2, 3)</code>
Create a tensor of ones	<code>torch.ones(2, 3)</code>
Create a tensor with random values	<code>torch.rand(2, 3)</code>
Create a tensor with a range	<code>torch.arange(0, 10, 2)</code>
Create an identity matrix	<code>torch.eye(3)</code>
Create a tensor from NumPy	<code>torch.from_numpy(np_array)</code>

2. Tensor Operations

Operation	Code Example
Element-wise addition	<code>tensor1 + tensor2</code> or <code>torch.add(tensor1, tensor2)</code>
Element-wise multiplication	<code>tensor1 * tensor2</code> or <code>torch.mul(tensor1, tensor2)</code>
Matrix multiplication	<code>torch.matmul(tensor1, tensor2)</code>
Transpose a tensor	<code>tensor.t()</code> or <code>torch.transpose(tensor, 0, 1)</code>
Reshape a tensor	<code>tensor.view(3, 4)</code> or <code>tensor.reshape(3, 4)</code>
Concatenate tensors	<code>torch.cat([tensor1, tensor2], dim=0)</code>
Stack tensors	<code>torch.stack([tensor1, tensor2], dim=0)</code>
Sum along a dimension	<code>torch.sum(tensor, dim=0)</code>
Mean along a dimension	<code>torch.mean(tensor, dim=0)</code>
Max along a dimension	<code>torch.max(tensor, dim=0)</code>
Min along a dimension	<code>torch.min(tensor, dim=0)</code>

3. Comparison Operators

Operation	Code Example
Less than	<code>torch.lt(tensor1, tensor2)</code> or <code>tensor1 < tensor2</code>
Less than or equal to	<code>torch.le(tensor1, tensor2)</code> or <code>tensor1 <= tensor2</code>
Greater than	<code>torch.gt(tensor1, tensor2)</code> or <code>tensor1 > tensor2</code>
Greater than or equal to	<code>torch.ge(tensor1, tensor2)</code> or <code>tensor1 >= tensor2</code>
Equal to	<code>torch.eq(tensor1, tensor2)</code> or <code>tensor1 == tensor2</code>
Not equal to	<code>torch.ne(tensor1, tensor2)</code> or <code>tensor1 != tensor2</code>

4. Indexing and Slicing

Operation	Code Example
Select a single element	<code>tensor[0, 0]</code>
Select a row	<code>tensor[0, :]</code>
Select a column	<code>tensor[:, 0]</code>
Slice a tensor	<code>tensor[0:2, 1:3]</code>
Masking with a condition	<code>tensor[tensor > 0.5]</code>

5. Tensor Manipulation

Operation	Code Example
Squeeze (remove size-1 dimensions)	<code>torch.squeeze(tensor)</code>

Operation	Code Example
Unsqueeze (add a dimension)	<code>torch.unsqueeze(tensor, dim=0)</code>
Permute dimensions	<code>tensor.permute(1, 0, 2)</code>
Flatten a tensor	<code>tensor.flatten()</code>
Clip values	<code>torch.clamp(tensor, min=0, max=1)</code>

6. Reduction Operations

Operation	Code Example
Sum all elements	<code>torch.sum(tensor)</code>
Mean of all elements	<code>torch.mean(tensor)</code>
Max of all elements	<code>torch.max(tensor)</code>
Min of all elements	<code>torch.min(tensor)</code>
Argmax (index of max value)	<code>torch.argmax(tensor)</code>
Argmin (index of min value)	<code>torch.argmin(tensor)</code>

7. Autograd and Gradients

Operation	Code Example
Enable gradient tracking	<code>tensor.requires_grad_(True)</code>
Compute gradients	<code>loss.backward()</code>
Access gradients	<code>tensor.grad</code>
Disable gradient tracking	<code>with torch.no_grad():</code>
Detach from computation graph	<code>tensor.detach()</code>

8. Neural Network Module

Operation	Code Example
Define a neural network	<code>class Net(nn.Module): ...</code>
Forward pass	<code>output = model(input)</code>
Loss function	<code>loss = nn.CrossEntropyLoss()</code>
Optimizer	<code>optimizer = torch.optim.SGD(model.parameters(), lr=0.01)</code>
Training loop	<code>for epoch in range(epochs): ...</code>

9. Device Management

Operation	Code Example
Move tensor to CPU	<code>tensor.cpu()</code>
Move tensor to GPU	<code>tensor.cuda()</code>
Check device	<code>tensor.device</code>
Set default device	<code>torch.device('cuda' if torch.cuda.is_available() else 'cpu')</code>

10. Saving and Loading Models

Operation	Code Example
Save a model	<code>torch.save(model.state_dict(), 'model.pth')</code>
Load a model	<code>model.load_state_dict(torch.load('model.pth'))</code>
Save entire model	<code>torch.save(model, 'model.pth')</code>
Load entire model	<code>model = torch.load('model.pth')</code>

11. Miscellaneous

Operation	Code Example
Set random seed	<code>torch.manual_seed(42)</code>

Operation	Code Example
Check tensor shape	tensor.shape or tensor.size()
Check tensor data type	tensor.dtype
Convert tensor to NumPy	tensor.numpy()
Convert NumPy to tensor	torch.from_numpy(np_array)

Example: Using Comparison Operators

```
python
import torch
```

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```
# Create tensors
```

```
tensor1 = torch.tensor([1, 2, 3])
```

```
tensor2 = torch.tensor([2, 2, 2])
```

```
# Comparison operations
```

```
print("Less than:", torch.lt(tensor1, tensor2))      # tensor([ True, False, False])
```

```
print("Less than or equal to:", torch.le(tensor1, tensor2)) # tensor([ True, True, False])
```

```
print("Greater than:", torch.gt(tensor1, tensor2))    # tensor([False, False,  True])
```

```
print("Greater than or equal to:", torch.ge(tensor1, tensor2)) # tensor([False, True,  True])
```

```
print("Equal to:", torch.eq(tensor1, tensor2))        # tensor([False,  True, False])
```

```
print("Not equal to:", torch.ne(tensor1, tensor2))    # tensor([ True, False,  True])
```